

Could Old Milton Parkway have preserved more trees?

A retrospective analysis of cost, traffic performance, and the evidence needed to assess tree-removal necessity.

Finding: The selected inward widening has a documented economic and construction rationale. The reviewed evidence does not establish that every removed tree had to be removed, or that no feasible design could preserve more mature canopy.

What the records show	What remains unresolved
A 42-foot typical median was reduced to 20 feet, with two additional 11-foot lanes.	Tree locations, root-zone impacts, and selective preservation feasibility.
Outside widening was evaluated and rejected as more costly and disruptive.	Net canopy preserved by outside or mixed widening, including exterior tree losses.
The cost premium was about \$12.6 million in total; about \$9.7 million was ROW.	Final designs, current alternative estimates, and the full environmental commitments.

Scope: GDOT PI 0017187, North Point Parkway to Kimball Bridge Road, Alpharetta, Georgia. The study uses historical concept estimates to evaluate the original choice. It does not estimate the cost of changing an active construction contract.

Student project status: Initial AI-assisted desk study and executable analysis completed. Fieldwork, canopy mapping, agency interviews, independent traffic calibration, and a final engineering preservation assessment have not been performed.

Sources: C = approved concept report (March 21, 2022); D = location/design report approved January 31, 2024. Page references throughout use PDF page numbers, not printed report numbers. Key pages are supplied in evidence_excerpt.pdf.

2. What was actually decided

The concept states that the broad median was originally intended to accommodate future expansion. Inward construction would retain much of the outside curb and pavement, add stormwater drainage in the median, and simplify traffic maintenance. This is the agency/design-team rationale, not independently verified design history. [C14]

Alternative	Total estimate	ROW estimate	Parcels	Months
Inside widening	\$29.197m	\$1.506m	18	30
Outside widening	\$41.800m	\$11.200m	63	42
North Point grade separation	\$32.500m	\$5.300m	15	48

Recalculated outside-minus-inside differences: \$12,603,205 total (+43.2%); \$9,694,000 ROW; 45 additional affected parcels; 12 additional estimated construction months. ROW accounts for 76.9% of the total premium. Values are historical estimates, not realized savings. [C14; analysis.py]

The narrative identifies drawbacks specific to its outside option: connecting existing right-turn lanes into through lanes, shoulder reconstruction, utility effects, and additional property/environmental impacts. It asserts a rear-end crash concern from losing turn lanes. That does not disprove every possible outward or hybrid design. [C14]

The inside option still anticipated 18 affected parcels, including five city-owned parcels, with no business or residential displacements. The lower number of grade-separation parcels does not imply lower land area or cost. [C10, C14]

Cost discipline: Use the same concept table for alternative comparisons. Do not subtract a 2026 construction estimate from a 2021-era total estimate. The concept summary includes some unresolved environmental mitigation costs, and the outside estimate is rounded. The arithmetic is exact to the listed inputs, but practical precision is about \$12.6m and \$9.7m. [C13-C14]

The January 2024 approval retains the inward-widening approach, identifies Alpharetta as responsible for property rights, and describes replacement of two existing bridges with one wider bridge. [D2-D5]

3. Geometry explains impact, not every removal

The typical-section arithmetic is $42 - (2 \times 11) = 20$ feet. The median loses 22 feet, or 52.4% of its typical width. This is a width calculation, not a measurement of canopy loss: intersections and the bridge differ, and the concept notes median widths ranging from 6 to about 40 feet in select places. [C6-C7]

Evidence	Interpretation
New lanes and much drainage lie in the median. [C12, C14]	Trees can conflict with excavation and grading even where their trunks are outside pavement.
A landscaped median was proposed. [C14]	A planting intention does not establish preservation of mature trees or replacement of their functions.
Median aesthetics were a frequent public-comment topic. [C12]	Tree/landscape concerns were present before construction; the individual responses still need review.
Final clearing and arborist records are not in this analysis.	No count of avoidable removals is defensible yet.

Three separate tests: (1) Is some capacity intervention justified? (2) Is this alignment the preferable way to deliver it? (3) Does a particular tree conflict with necessary work? A yes to the first does not automatically answer the other two.

A localized preservation option should be screened against pavement, turn-lane storage, sight distance, drainage, utility excavation, grading, root survival, bridge approaches, and construction staging. Unknown feasibility cannot be scored as feasible.

Outside widening could spare median trees while removing trees on adjacent land. The appropriate ecological comparison is net additional surviving canopy across the entire affected footprint. A replacement sapling is not immediately equivalent to a mature tree.

Current conclusion: Substantial median disturbance is consistent with the chosen design. Universal necessity of tree removal remains unproven by the reviewed evidence. This is an evidence limit, not proof of an error by the project team.

4. Traffic benefits are uneven

The project justification reports modeled corridor travel-time reductions of 2 minutes in 2030 and 2.7 minutes in 2050. It reports corridor LOS D for the 2030 build case but LOS E by 2050. Those are historical forecasts, not measured outcomes. The concept also reports a 2.98 operations/safety benefit-cost ratio, which this study has not independently replicated. [C27, C14]

The table below covers all seven named signalized junctions, using Attachment 8 opening-year values. Change = build minus no-build; positive values indicate more delay. The concept opening year was 2027, not the current anticipated completion year. [C5, C9, C129]

Intersection	AM: no-build to build	Change	PM: no-build to build	Change
North Point Parkway	31.0 to 34.2	+3.2	40.4 to 35.5	-4.9
Cotton Creek / Brookside West	6.4 to 2.2	-4.2	6.4 to 5.7	-0.7
Vista Forest / Brookside East	7.7 to 3.5	-4.2	6.3 to 13.5	+7.2
Park Bridge / Alexander	20.9 to 3.6	-17.3	8.7 to 4.7	-4.0
Parkview / Southbridge	9.3 to 11.3	+2.0	9.5 to 10.8	+1.3
State Bridge Way	2.8 to 6.7	+3.9	2.5 to 1.8	-0.7
Kimball Bridge Road	38.5 to 51.0	+12.5	53.0 to 53.0	+0.0

Delay units are interpreted as seconds per vehicle, consistent with the capacity/LOS outputs; the summary page itself labels the column only "Delay." Do not mix worst-movement V/C with intersection-average delay: the page explicitly labels V/C as the worst movement. [C129]

Examples: North Point PM improves by 4.9 seconds, while Kimball Bridge AM worsens by 12.5 seconds. These comparisons show distributional effects; they do not invalidate a corridor travel-time benefit. Summing these intersection averages would not yield a valid corridor travel-time estimate without movement flows and routes.

Validation needed: Reconcile the concept summary and justification forecast values, recover the native traffic models, and test the final design against updated demand and signal timing. An optimized four-lane option cannot be declared sufficient from this desk study. [C5 versus C27]

5. A preservation break-even model

Rather than invent a tree count or assign an arbitrary environmental value, calculate the extra cost per additional surviving mature tree. This is a screening threshold, not a valuation of trees or a benefit-cost conclusion.

Threshold = incremental cost / net additional surviving trees. Use the same time horizon and price year. The denominator must include losses outside the median and survival risk. Zero or negative additional survival gives no positive preservation benefit.

Hypothetical net additional trees	Outside premium per tree
50	\$252,064
100	\$126,032
200	\$63,016
400	\$31,508

The table uses the documented \$12.603m historical outside premium. All tree counts are hypothetical; none were observed. With 200 additional survivors, the threshold is about \$63,016 per tree before accounting for other benefit/cost differences.

A localized alternative costing a hypothetical \$1m and retaining 100 additional trees would have a \$10,000-per-tree threshold. No such alternative has yet been designed or costed. This demonstrates why selective preservation deserves a separate screen rather than assuming the only choice is \$12.6m of full outside widening.

analysis.py also computes a grid using \$1m, \$3m, \$6m, and \$12.603m premiums across 50, 100, 200, and 400 trees. These inputs are sensitivity scenarios, not estimates of what Alpharetta should spend.

Multi-objective decision: First exclude alternatives that fail engineering requirements. Then compare cost, net canopy survival, traffic delay, affected property, and disruption. An option is dominated only if another is no worse on every relevant measure and better on at least one. Missing canopy data prevents a full dominance conclusion.

Avoid double counting: If environmental benefits are later monetized, distinguish replacement cost, carbon storage, annual sequestration, runoff benefits, and amenity values. Do not add overlapping benefit estimates. Keep nonmonetized outcomes visible.

6. Project plan and remaining evidence

Phase / status	Work product and completion test
1. Decision audit / completed	Source-linked findings, historic cost reconciliation, seven-junction comparison, and executable sensitivity analysis.
2. Canopy audit / pending records	Final clearing and drainage plans over dated preconstruction imagery; median/exterior canopy mapped separately, with uncertainty.
3. Selective preservation / pending screening	One or two candidate clusters reviewed for roots, geometry, utilities, drainage, traffic, and cost. No assumed feasibility.
4. Review and communicate / pending	Agency corrections, revised findings, reproduced calculations, and a student-authored brief.

The accompanying project_plan.md provides the detailed workflow, estimated student effort, acceptance criteria, and a ready-to-send records request. No request has been sent. Priority records are final clearing/grading/utility plans, the tree inventory or arborist assessment, and the approved environmental document and commitments.

The 2024 report records a categorical exclusion approval dated January 19, 2024. Its referenced environmental document has not been reviewed here. The report also identifies a July 2, 2021 response letter. Its stated public-involvement date range appears inconsistent with that letter and the 2022 concept; retain the discrepancy for agency clarification. [D2; C12]

Most useful next question: For trees removed from the retained median area, what documented conflict required removal, and was selective preservation evaluated? A plan-sheet response to that question would advance the study more than a general explanation that the road is being widened.

Student contribution: The research and draft model were generated with AI assistance. Before describing this as your project, examine the source packet, run the model, and add your own documented analysis, mapping, or correspondence. Do not claim field measurements, implementation, or independently authored work that has not occurred.

A defensible eventual resume description: Investigated SR 120 widening tradeoffs using public engineering records; reproduced alternative-cost comparisons and evaluated mature-tree preservation scenarios through sensitivity analysis. Tailor it to work you actually understand and perform.

7. Sources, audit trail, and limitations

C. GDOT / City of Alpharetta / AECOM. Approved concept report, March 21, 2022; 680-page package. [Download original report](#). Reviewed core pages: 3-7, 9-15, 27, 128-131. Key source: PDF p14 = printed report p13. The package was downloaded and its text extracted; not all scanned attachments were visually reviewed.

D. GDOT. Approved location and design report, January 31, 2024; five pages. [Download original report](#). Pages 2-3 confirm design and environmental approval references; page 5 provides the public-inspection notice and historical project contact.

G. GDOT GeoPI PI 0017187. Retrieved through the search index September 5, 2026; direct webpage retrieval was unreliable. Lists length 1.9508 miles and the report download links. Used for discovery and scope, not to replace concept-era alternative costs.

A. [City construction announcement, April 2026](#). Describes utility clearing and anticipated late-summer 2029 completion. Schedules can change; the study does not independently verify construction progress.

Evidence packet pages	Original PDF pages
1-11	C3, C4, C5, C6, C7, C10, C12, C13, C14, C27, C129
12-15	D2, D3, D4, D5

Reproduction: Run python analysis.py in the outputs folder using standard Python 3. It writes analysis_results.json and checks cost deltas, delay deltas, and rejection of a zero-tree denominator. Inputs are embedded with page references. No proprietary software or network connection is required.

Limitations: No pre/post canopy estimate; no tree count; no engineering certification of an alternative; no current traffic observation; no independent B/C reconstruction; no final plan verification. Public-response counts are not a representative opinion survey. The original \$12m ROW claim is corrected, but no original city statement containing that figure was located.

Answer to the motivating question: The records explain why the city and GDOT selected inward widening. They do not establish whether all mature-tree losses were unavoidable. That narrower question requires the final disturbance footprint and a tree-survival assessment.